

Introduction & State of the Art

UniMentor / Software Engineering · Problem context, SMART goals, and comparative market analysis against industry competitors.

1.1 Problem Context

The environment

At a university, academic support is spread across many informal channels: department notice boards, peer-tutoring programmes run by individual faculties, personal contacts, class group chats and email. A student who is struggling with a subject such as Calculus, Algorithms or Thermodynamics usually does not know whether a tutor exists, who it is, what that tutor is good at, or when they are free. A tutor or mentor, on the other hand, has no single place to say "I am available on Tuesdays from 16:00 to 18:00 for Algorithms" and no way to see who has actually committed to a slot.

The process today typically looks like this:

1. The student hears about a tutor by word of mouth or finds a poster.
2. The student writes an email or message and waits for an answer.
3. The two negotiate a time through several back-and-forth messages.
4. The session happens (or not, if one side forgets). Nothing is recorded.
5. Neither side keeps a history of what was covered or whether it helped.

Why the current process is inefficient

Inefficiency	Consequence
No discoverability. Tutors and their areas of expertise are not listed anywhere central.	Students give up, or pick the wrong tutor for their need.
Manual scheduling. Availability is negotiated by message.	Many messages for one booking; slots are double-booked or missed because nothing enforces the schedule.
No reminders or confirmations.	No-shows and forgotten sessions waste the tutor's time.
No record of sessions.	There is no history, no feedback and no way to track a student's progress over a semester.

Inefficiency	Consequence
No visibility for the institution.	Nobody can tell how much tutoring is used, so educational resources are underused and cannot be improved.

Who suffers from the problem

- **Students** (primary). They lose time and often do not get help when they need it, which affects grades and drop-out risk. First-year students and students new to the university are hit hardest because they have no network.
- **Tutors and mentors.** They are willing to help but spend their effort coordinating instead of teaching, and they get no recognition because their work is not recorded.
- **Programme administrators.** They cannot measure demand, supply or outcomes.
- **The university.** Existing educational resources are underused.

Proposed response

UniMentor is a web platform that centralises the whole cycle: a tutor publishes expertise and availability, a student finds a suitable tutor and books a session in an interactive calendar without conflicts, both receive email confirmations and reminders, and after the session notes and feedback are stored in a personal history.

When booking, the student chooses one of three **session types**, so that the length of the session matches the need:

Session type	Maximum duration	Purpose
Quick Check	5 minutes	A short conversation in which the tutor asks the student about their current academic performance and detects problems early.
Standard Tutoring	30 minutes	Solving doubts about exercises and course material.
Deep Dive	1 hour	Solving deeper doubts that need more time, such as a whole topic or exam preparation.

Availability is published in fixed 60-minute blocks (BR-CONF-03, Section 2), and every booking reserves one whole block. The session type chosen at booking sets the intended maximum length of the conversation inside that block, so the tutor knows what to prepare. This gives students a low-effort way to start (a five-minute check) and a path to longer help, and it lets tutors manage their time in predictable blocks. (A Quick Check still reserves a full block; shorter slots are a possible future improvement.) To keep the project manageable, video calls (external Meet/Zoom links), payments and real-time chat are explicitly out of scope.

1.2 SMART Goals and KPIs

The five goals below refine the three project objectives of the proposal (search and booking, expertise profiles, session records) and add two quality goals (reliability and performance). Each goal is Specific, Measurable, Achievable, Relevant and Time-bound. Targets marked as *pilot* assume a small pilot at one university; the numbers should be validated by the team.

#	Goal	KPI	Target	Measured by	Deadline
G1	Fast booking. A student can find a tutor, choose a session type (Quick Check, Standard or Deep Dive) and book it without any external messaging.	Time from landing on the search page to a confirmed booking	≤ 3 minutes (median) and ≤ 6 clicks	Usability test with 5 students, stopwatch plus click count	End of first functional release
G2	No schedule conflicts. The system never lets two sessions overlap for the same tutor or the same student.	Double bookings produced	0 in 100 % of automated concurrency tests (50 simultaneous requests for the same time yield exactly 1 booking)	Automated concurrency test on the booking endpoint	Before first release
G3	Good tutor-student matching. Tutors describe their expertise so students can search by subject.	% of tutor profiles with at least 1 subject and a short description; % of searches by subject that return at least one tutor	≥ 90 % complete profiles; ≥ 80 % of subject searches non-empty (<i>pilot</i>)	Database query plus search logs	4 weeks after pilot start
G4	Centralised session record. Completed sessions carry notes and feedback.	% of completed sessions with at least a tutor note or a student rating	≥ 70 % (<i>pilot</i>)	Database query	End of pilot semester

#	Goal	KPI	Target	Measured by	Deadline
G5	Reliable and responsive service. The system answers fast and notifies on time.	(a) 95th-percentile latency of availability and booking endpoints; (b) share of confirmation emails delivered within 60 s; (c) share of reminders sent 24 h $\pm$ 15 min before the session	(a) < 200 ms at 50 concurrent users; (b) $\geq$ 95 %; (c) $\geq$ 95 %	Load test (e.g. k6) and email-log analysis	End of first functional release

Why these numbers

- G1 compares with the current process, which takes at least one message round-trip (hours or days). Three minutes is realistic for a calendar-based flow.
- G2 is the most important technical goal because the proposal names collision handling as the main risk. Its target is absolute (zero).
- G5(a) uses 200 ms because it is the usual threshold at which an interface feels instant. 50 concurrent users is far above what a single-university pilot needs (assumption, to be validated with the university tutoring programme before the pilot).
- G3 and G4 depend on user behaviour, so they are set as adoption targets for a pilot rather than technical guarantees.

1.3 State of the Art Analysis

Method

We compared UniMentor with two products that cover different parts of the problem. Information comes from the vendors' public pages and third-party reviews (see References). Where a vendor does not publish a fact (mostly technology stack and internal scalability), we say so instead of guessing.

- **Calendly** is a general-purpose scheduling tool. It represents the "calendar" half of our problem.
- **Superprof / Wyzant** are tutoring marketplaces. We use **Wyzant** as the second reference because its fees are clearly documented, and mention Superprof as additional context.

Comparative matrix

Dimension		(tutoring marketplace)	(proposed)
Primary purpose	Scheduling meetings through booking links	Marketplace where students hire private tutors	University-run tutoring management
Calendar and booking	Yes: availability rules, calendar sync, reminders, group events	Booking and scheduling within the platform	Yes: tutor availability, student booking, conflict prevention
Session types	Several event types with different durations on paid plans (the free plan allows one); durations are generic, not tied to tutoring purposes	Lesson length is agreed per tutor and billed hourly; no purpose-based types	Three predefined types by purpose (Quick Check 5 min, Standard 30 min, Deep Dive 60 min), booked inside fixed 60-minute blocks
Tutor/mentor profiles by expertise	No, it is not a tutoring product; there is no subject search	Yes: profiles, subjects, ratings, search	Yes: profile with areas of expertise, searchable by subject
Session notes, feedback and progress	No (meeting scheduling only)	Session reports and reviews; reports exist mainly for billing	Yes: notes, feedback and history are a core feature
Roles	Hosts and invitees; team roles on paid plans	Students and tutors	Student, Tutor, Admin (university oversight)
Notifications	Email and SMS reminders on paid plans	Email and in-platform messaging	Email confirmations and reminders
Payments	Optional via Stripe/PayPal on paid plans	Mandatory: platform takes payment from students	None, assumed free university service (out of scope)
Cost to the user	Free tier limited to 1 event type; Standard about \$10–12 per seat per month; Teams about \$16–20 per seat per month; Enterprise from about \$15,000 per year	Tutor pays a flat 25 % fee; student pays about 9 % service fee on top of the tutor's rate	Free for students and tutors; cost is hosting only (free/low tiers of Vercel and Render)

Dimension		(tutoring marketplace)	(proposed)
Scalability	High: commercial SaaS with a public API and webhooks; scale is the vendor's responsibility	High: commercial marketplace, operated by the vendor	Modest by design: a single university; a layered monolith can scale vertically and later be split
Data ownership and integration	Vendor-hosted; integrates with many tools through API	Vendor-hosted; no university integration	Owned by the university; can be adapted to its needs
Technology stack	React frontend embedded in a Ruby on Rails monolith, with separately built React microfrontends (Webpack Module Federation), per Calendly's own engineering blog	Not publicly documented by the vendor; no verified Wyzant-only stack	HTML/CSS/JavaScript, Node.js + Express, PostgreSQL, Vercel and Render

Analysis

What the competitors do well. Calendly has a mature booking experience and a robust API. Wyzant has the matching, reviews and trust features a tutoring service needs.

Gaps that justify UniMentor.

1. *Calendly has no tutoring concept.* No expertise profiles, no subject search, no session history, and meaningful features require paid seats, which a university programme would have to fund per tutor.
2. *Wyzant is commercial.* It is built around payments and fees (25 % plus about 9 %), is open to anyone, and is not integrated with a university's students, tutors or oversight. It is unsuitable for a free, institution-run service.
3. *Neither tracks academic progress over time inside an institution.* Session notes and feedback tied to a student's university record are not a feature of either product.
4. *Neither gives the institution oversight.* An Admin role that sees demand and usage is missing.

Where UniMentor is weaker, honestly. We will not match the scale, polish or ecosystem of commercial tools, and we must build our own calendar logic, which is the project's main technical risk. We accept this trade-off because the scope is deliberately small and fits a single institution.

Positioning. UniMentor combines Calendly's booking convenience with a marketplace's expertise matching, in a free, institution-owned product with session records.

References

- Calendly pricing and plans, 2026: [Zeeg](#), [Jotform](#), [Koalendar](#)
- Calendly technology stack (first-party engineering article, 30 Sep 2026): [Microfrontends Need a Platform, Not Just a Boundary](#)
- Wyzant fees: [Wyzant support, service fee charged to students](#), [TutorTab](#), [Wyzant commission](#)
- Superprof model and fees: [Superprof, how it works](#), [Superprof payment and commission](#)

Note: Prices are third-party reports checked in 2026 and may change; re-verify before the final submission.